GABRIEL-ROSENBERG RECONSTRUCTION: EQUIVALENCES OF QCOH COME FROM ISOMORPHISMS OF SCHEMES AND INVERTIBLE MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original reference 01PH: sources/stacks-chow/tags.txt, source lines 2230–2230. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GPE: sources/stacks-chow/tags.txt, source lines 20824–20824. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GPG: sources/stacks-chow/tags.txt, source lines 20826–20826. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GPH: sources/stacks-chow/tags.txt, source lines 20827–20827. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GPI: sources/stacks-chow/tags.txt, source lines 20828–20828. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GP0: sources/stacks-chow/tags.txt, source lines 20810–20810. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference shared-bibliography-key: sources/stacks-functors/SOURCE.json, source lines 14–15. Author's bibliographic citation inside the counted span (reference environment pointing at Brandenburg); the shared Stacks bibliography is not staged locally, so the cite key is declared here per SOURCE.json bibliography_note. Pointer only; its content is not part of this item.

Proposition 0.1. *Let X and Y be quasi-compact and quasi-separated schemes. If $F : QCoh(\mathcal{O}_X) \to QCoh(\mathcal{O}_Y)$ is an equivalence, then there exists an isomorphism $f : Y \to X$ of schemes and an invertible $\mathcal{O}_Y$-module $\mathcal{L}$ such that $F(\mathcal{F}) = f^*\mathcal{F} \otimes \mathcal{L}$.*

Date: stacks-project:functors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Of course F is additive, exact, commutes with all limits, commutes with all colimits, commutes with direct sums, etc. Let $U \subset X$ be an affine open subscheme. Let $\mathcal{I} \subset \mathcal{O}_X$ be a finite type quasi-coherent sheaf of ideals such that $Z = V(\mathcal{I})$ is the complement of U in X, see Properties, Lemma 01PH. Then $\mathcal{O}_X/\mathcal{I}$ is a finitely presented $\mathcal{O}_X$-module. Hence $\mathcal{G} = F(\mathcal{O}_X/\mathcal{I})$ is a finitely presented $\mathcal{O}_Y$-module by Lemma 0GPE. Denote $T \subset Y$ the support of $\mathcal{G}$ and set $V = Y \setminus T$. Since $\mathcal{G}$ is of finite presentation, the scheme V is a quasi-compact open of Y. By Lemma 0GPG we see that F induces an equivalence between

(1) the full subcategory of $QCoh(\mathcal{O}_X)$ consisting of modules supported on Z, and
(2) the full subcategory of $QCoh(\mathcal{O}_Y)$ consisting of modules supported on T.

By Lemma 0GPH we obtain a commutative diagram

$$\begin{array}{ccc} QCoh(\mathcal{O}_X) & \xrightarrow[F]{} & QCoh(\mathcal{O}_Y) \\ \downarrow & & \downarrow \\ QCoh(\mathcal{O}_U) & \xrightarrow{F_U} & QCoh(\mathcal{O}_V) \end{array}$$

where the vertical arrows are the restruction functors and the horizontal arrows are equivalences. By Lemma 0GPI we conclude that V is affine. For the affine case we have Lemma 0GP0. Thus we find that there is an isomorphism $f_U : V \to U$ and an invertible $\mathcal{O}_V$-module $\mathcal{L}_U$ such that F_U is the functor $\mathcal{F} \mapsto f_U^*\mathcal{F} \otimes \mathcal{L}_U$.

The proof can be finished by noticing that the diagrams above satisfy an obvious compatibility with regards to inclusions of affine open subschemes of X. Thus the morphisms f_U and the invertible modules $\mathcal{L}_U$ glue. We omit the details. □